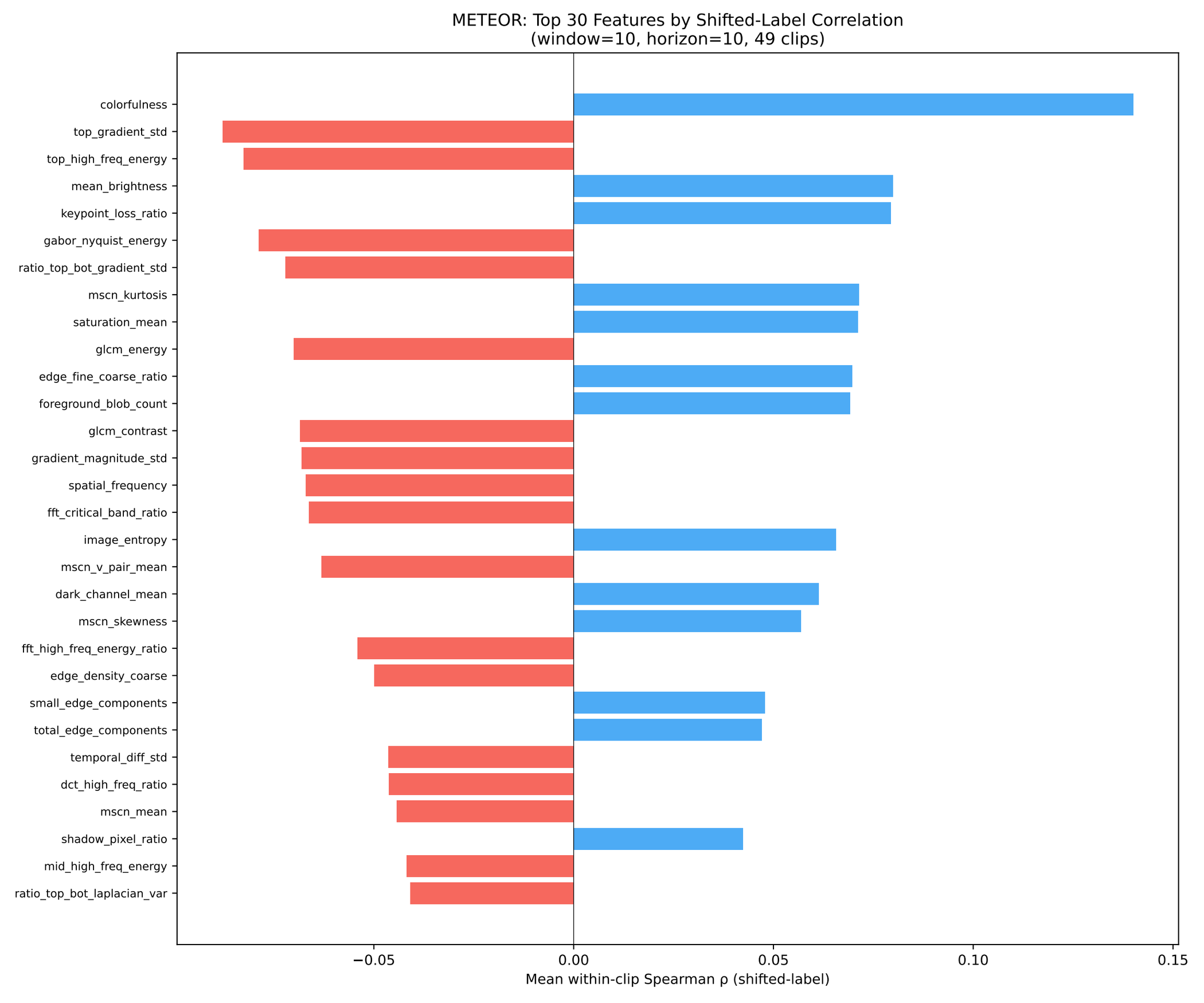
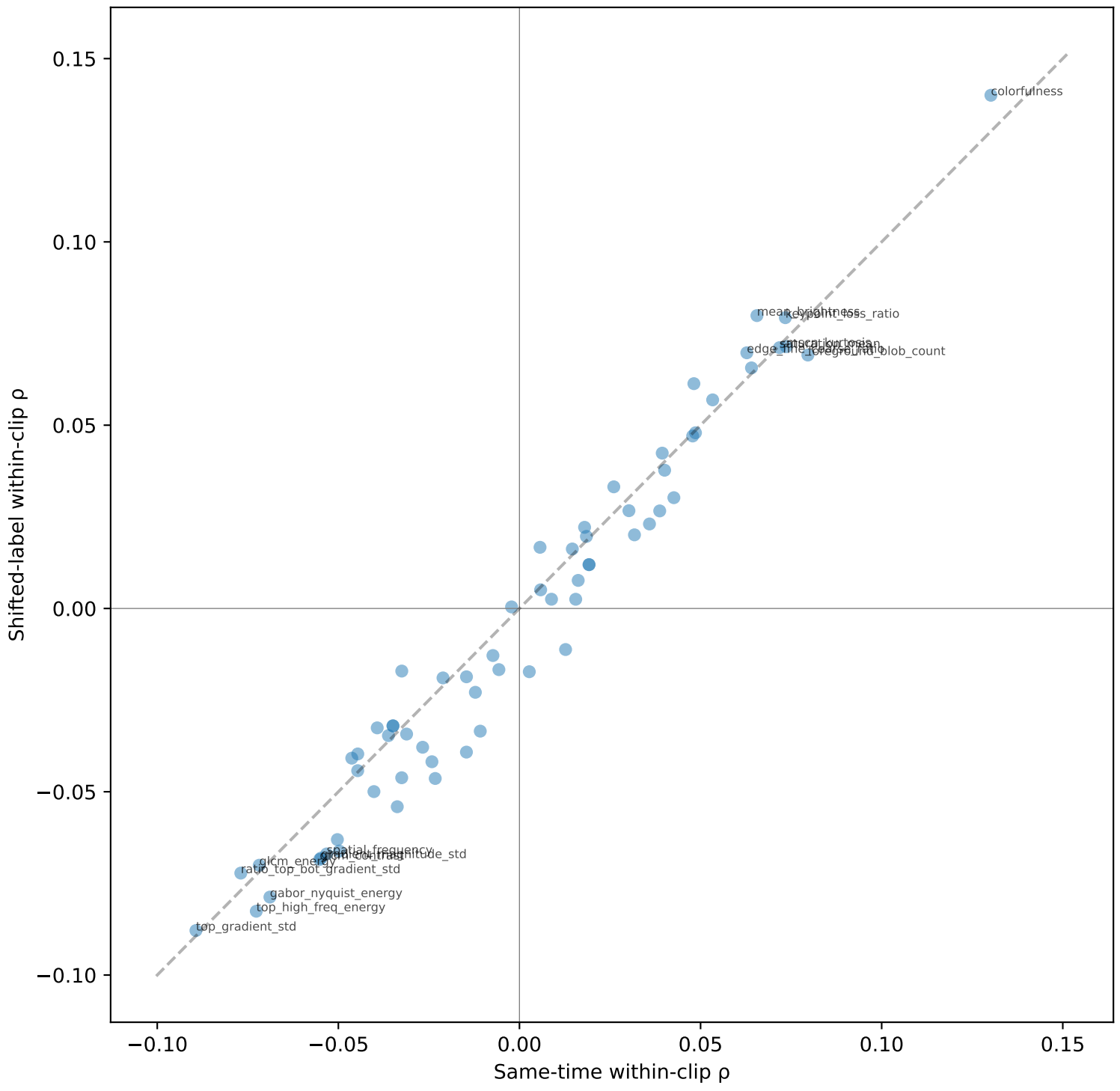
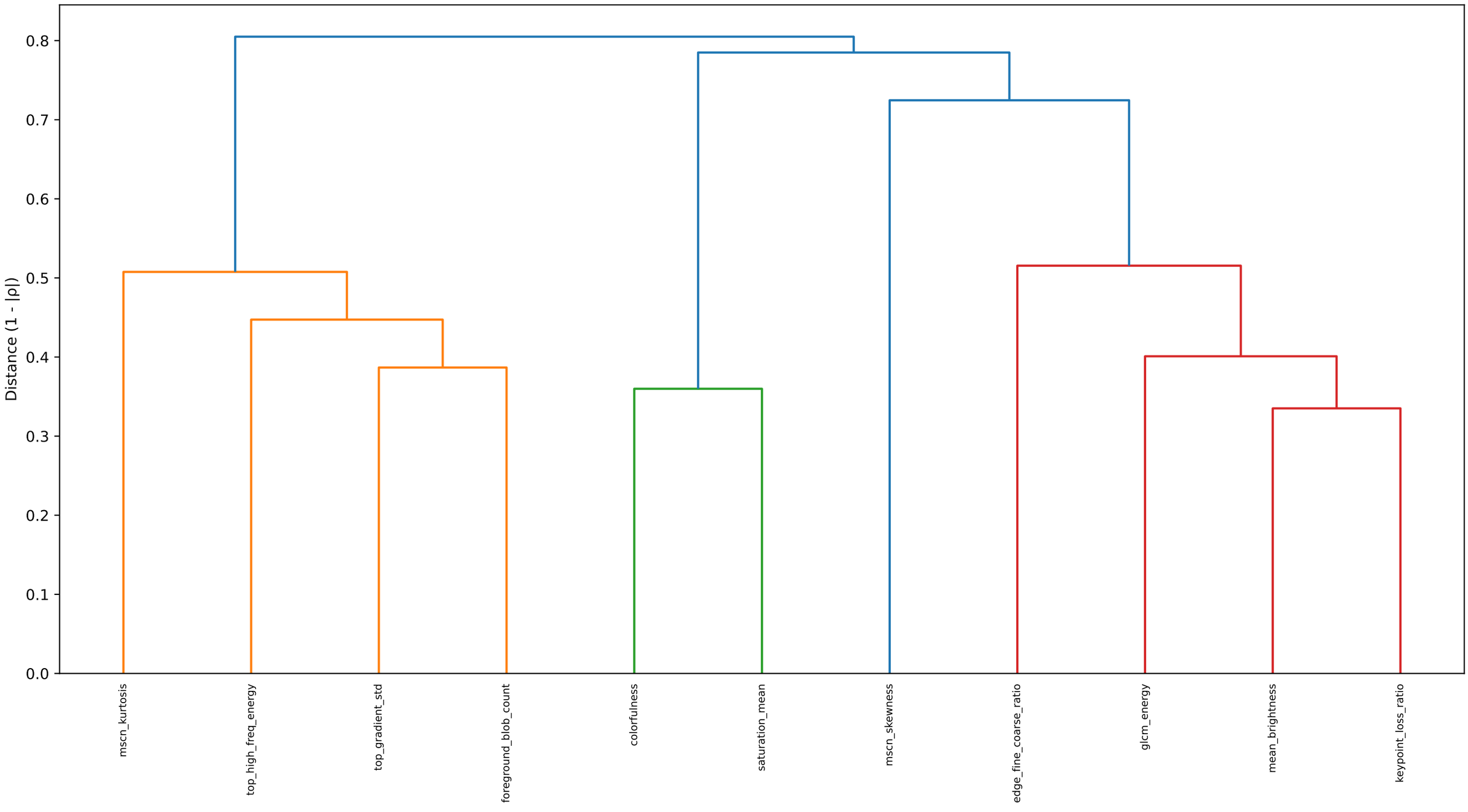
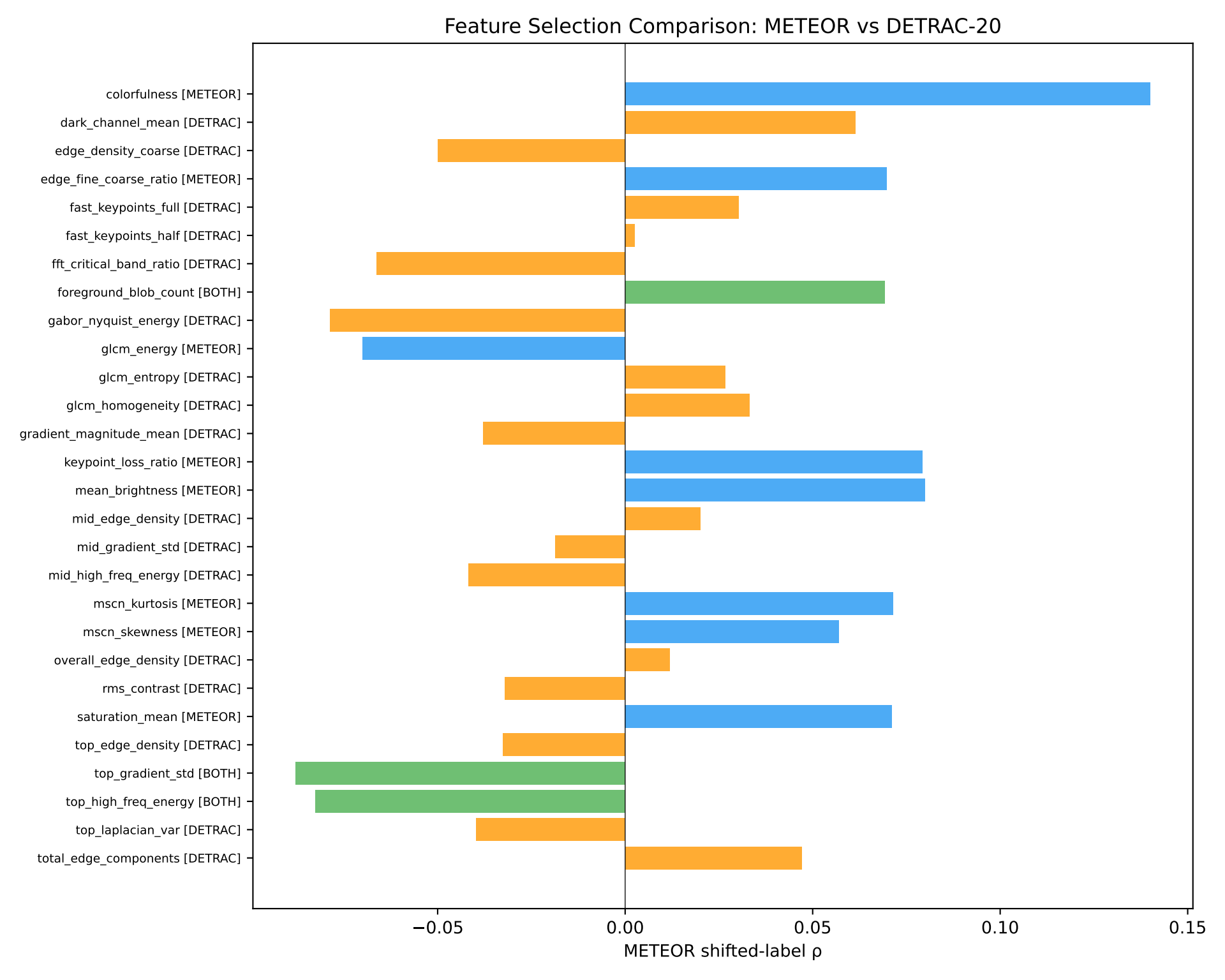

METEOR: Top 30 Features by Shifted-Label Correlation
(window=10, horizon=10, 49 clips)
colorfulness
top_gradient_std
top_high_freq_energy
mean_brightness
keypoint_loss_ratio
gabor_nyquist_energy
ratio_top_bot_gradient_std
mscn_kurtosis
saturation_mean
glcm_energy
edge_fine_coarse_ratio
foreground_blob_count
glcm_contrast
gradient_magnitude_std
spatial_frequency
fft_critical_band_ratio
image_entropy
mscn_v_pair_mean
dark_channel_mean
mscn_skewness
fft_high_freq_energy_ratio
edge_density_coarse
small_edge_components
total_edge_components
temporal_diff_std
dct_high_freq_ratio
mscn_mean
shadow_pixel_ratio
mid_high_freq_energy
ratio_top_bot_laplacian_var
−0.05
0.00
0.05
0.10
0.15
Mean within-clip Spearman ρ (shifted-label)


METEOR: Same-Time vs Shifted-Label Correlations
Same-time within-clip ρ
Shifted-label within-clip ρ
0.15
0.10
0.05
0.00
−0.05
−0.10
−0.10
−0.05
0.00
0.05
0.10
0.15
colorfulness
mean_brightness
keypoint_loss_ratio
edge_density
foreground_blob_count
spatial_frequency
gicm_energy
ratio_top_bot_gradient_std
gabor_nyquist_energy
top_high_freq_energy
top_gradient_std


METEOR: Dendrogram of 11 Selected Features
Distance (1 - |ρ|)
0.0
0.1
0.2
0.3
0.4
0.5
0.6
0.7
0.8
mscn_kurtosis
top_high_freq_energy
top_gradient_std
foreground_blob_count
colorfulness
saturation_mean
mscn_skewness
edge_fine_coarse_ratio
glcm_energy
mean_brightness
keypoint_loss_ratio


Feature Selection Comparison: METEOR vs DETRAC-20
colorfulness [METEOR]
dark_channel_mean [DETRAC]
edge_density_coarse [DETRAC]
edge_fine_coarse_ratio [METEOR]
fast_keypoints_full [DETRAC]
fast_keypoints_half [DETRAC]
fft_critical_band_ratio [DETRAC]
foreground_blob_count [BOTH]
gabor_nyquist_energy [DETRAC]
glcm_energy [METEOR]
glcm_entropy [DETRAC]
glcm_homogeneity [DETRAC]
gradient_magnitude_mean [DETRAC]
keypoint_loss_ratio [METEOR]
mean_brightness [METEOR]
mid_edge_density [DETRAC]
mid_gradient_std [DETRAC]
mid_high_freq_energy [DETRAC]
mscn_kurtosis [METEOR]
mscn_skewness [METEOR]
overall_edge_density [DETRAC]
rms_contrast [DETRAC]
saturation_mean [METEOR]
top_edge_density [DETRAC]
top_gradient_std [BOTH]
top_high_freq_energy [BOTH]
top_laplacian_var [DETRAC]
total_edge_components [DETRAC]
−0.05
0.00
0.05
0.10
0.15
METEOR shifted-label ρ

```
METEOR 65-Feature Correlation Analysis
==================================================

Dataset: 27574 frames, 49 clips
Window: 10 frames, Horizon: 10 frames
Valid clips for correlation: 49

Clustering: threshold=0.3 (|ρ| > 0.7)
  → 17 clusters from 65 features

Selected features: 11
  Best: colorfulness (ρ=+0.1400)

Overlap with DETRAC-20: 3/11
  Shared: foreground_blob_count, top_gradient_std, top_high_freq_energy
  METEOR-only: 8
  DETRAC-only: 17
```